

SET 2: CUSTOMER FEEDBACK ANALYSIS

Dataset: Customer Satisfaction

Department	AIDS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R
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Date	29 September 2026
Day	Tuesday

Dataset: Customer Satisfaction

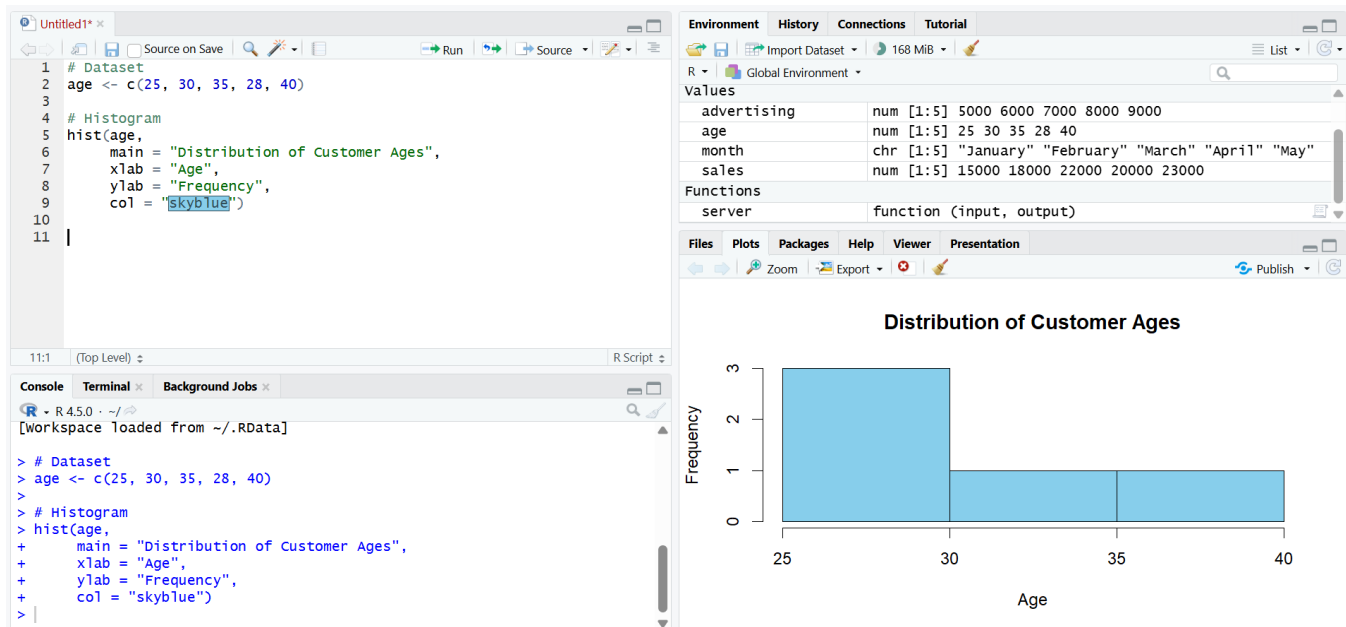
Customer ID	Age	Satisfaction Score
1	25	4
2	30	5
3	35	3
4	28	4
5	40	5

PROGRAM 1: Histogram of Customer Ages

Create a histogram to represent the distribution of customer ages. Label the axes and title the chart.

```
# Dataset
age <- c(25, 30, 35, 28, 40)

# Histogram
hist(age,
      main = "Distribution of Customer Ages",
      xlab = "Age",
      ylab = "Frequency",
      col = "skyblue")
```



Output 1: Distribution of Customer Ages

Observation: The histogram represents the distribution of the five customer ages in the given dataset.

PROGRAM 2: Pie Chart of Satisfaction Scores

Generate a pie chart to display the overall distribution of customer satisfaction scores and include labels.

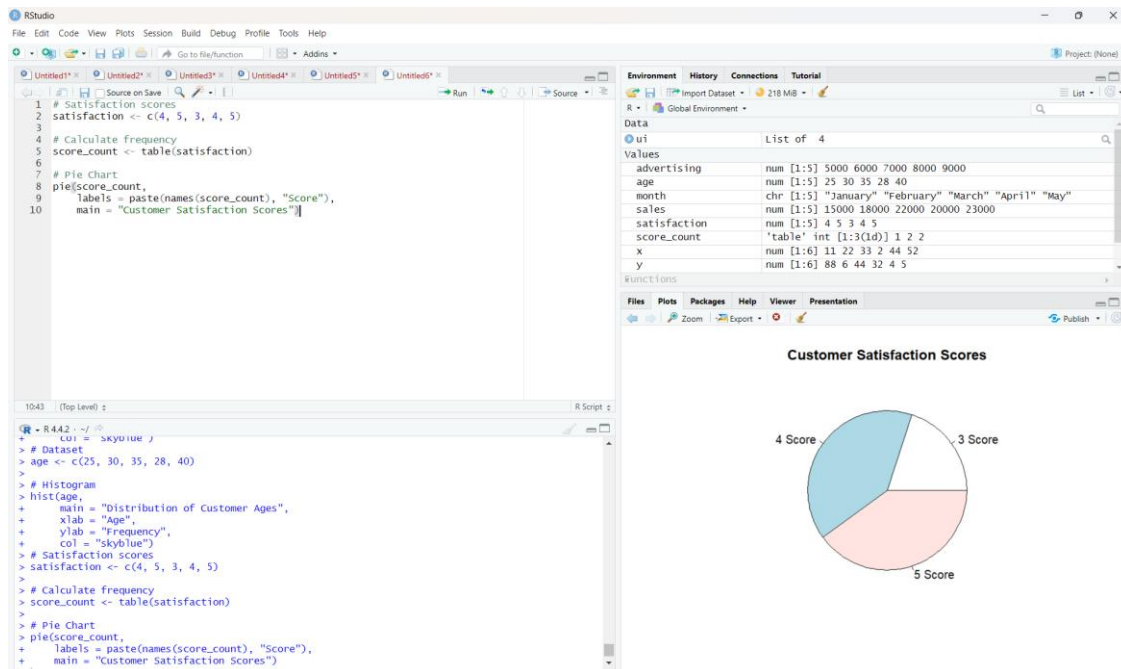
```

# Satisfaction scores
satisfaction <- c(4, 5, 3, 4, 5)

# Calculate frequency
score_count <- table(satisfaction)

# Pie Chart
pie(score_count,
    labels = paste(names(score_count), "Score"),
    main = "Customer Satisfaction Scores")

```



Output 2: Customer Satisfaction Scores Pie Chart

Observation: Scores 4 and 5 each occur twice, while score 3 occurs once.

PROGRAM 3: Stacked Bar Chart by Age Group

Build a stacked bar chart to visualize the distribution of customer satisfaction scores by age group.

```

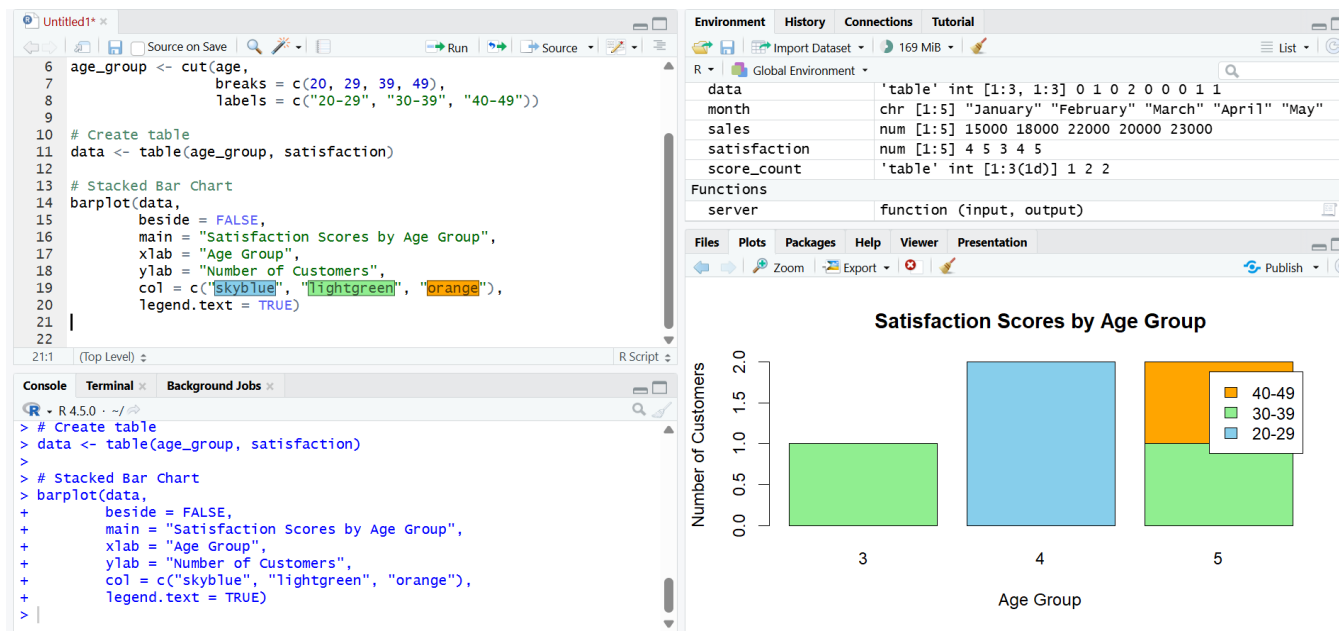
# Dataset
age <- c(25, 30, 35, 28, 40)
satisfaction <- c(4, 5, 3, 4, 5)

# Create age groups
age_group <- cut(age,
                 breaks = c(20, 29, 39, 49),
                 labels = c("20-29", "30-39", "40-49"))

# Create table
data <- table(age_group, satisfaction)

# Stacked Bar Chart
barplot(data,
        beside = FALSE,
        main = "Satisfaction Scores by Age Group",
        xlab = "Age Group",
        ylab = "Number of Customers",
        col = c("skyblue", "lightgreen", "orange"),
        legend.text = TRUE)

```



Output 3: Satisfaction Scores by Age Group

Observation: The code creates three age groups and stacks the satisfaction-score frequencies for each group.

Note: The supplied RStudio screenshot is inserted exactly as captured, including its displayed x-axis labels.

PROGRAM 4: Word Cloud of Customer Feedback

Develop a word cloud from open-ended customer feedback to identify prevalent customer sentiments.

```

library(wordcloud)
library(RColorBrewer)

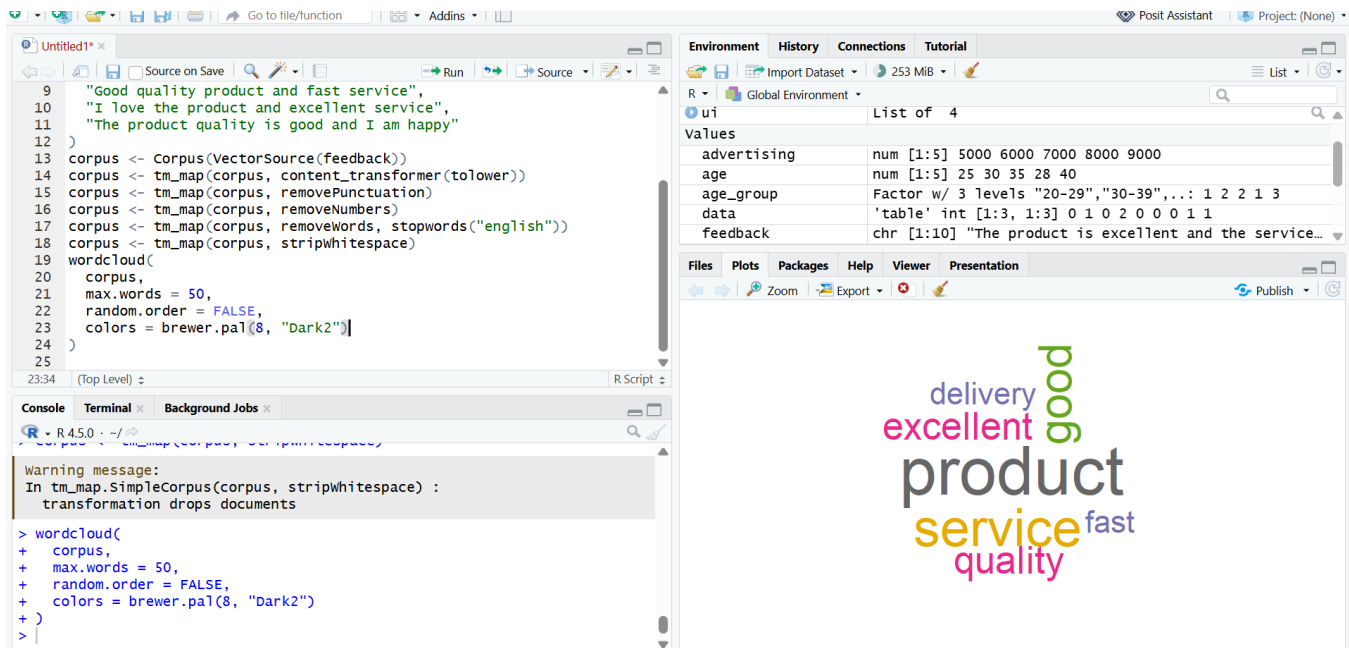
feedback <- c(
  "good service",
  "excellent service",
  "good experience",
  "friendly staff",
  "excellent service"
)

words <- unlist(strsplit(tolower(feedback), " "))

word_count <- table(words)

wordcloud(names(word_count),
  freq = word_count,
  min.freq = 1,
  scale = c(3, 0.5),
  colors = brewer.pal(8, "Dark2"),
  random.order = FALSE)

```



Output 4: Customer Feedback Word Cloud

Observation: The word cloud highlights frequently occurring terms in the supplied sample feedback, including service, excellent, good, friendly, and experience.

Procedure

1. Create vectors for customer age and satisfaction score.
2. Use `hist()` to represent the distribution of customer ages.
3. Use `table()` and `pie()` to display the satisfaction score distribution.
4. Create age groups using `cut()`, form a frequency table, and use `barplot()` for the stacked bar chart.
5. Prepare open-ended feedback text, count the words, and use `wordcloud()` to visualize frequent terms.

Result

The customer satisfaction data was successfully visualized using a histogram, pie chart, stacked bar chart, and word cloud.